

PRAKTIKUM SISTEM OPERASI

MODUL X

SHELL



Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

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JURNAL

1. Soal 1

Akan dimodifikasi shell dengan modifikasi syscall bernama chname yang berfungsi untuk mengubah nama suatu proses. Lihat kembali modul sebelumnya cara membuat syscall. Perhatikan sekarang syscall chname mempunyai 3 parameter yaitu pid, character dan panjang character. Character untuk menyimpan nama dan panjang character untuk panjang nama.

- a. Pada prototypes.h chname diubah menjadi:

```
/* in file chname.c */  
|  
extern pri16 chname(pid32, char *, uint32);
```

- b. Pada chname.c fungsi diubah dari:

```
pri16 chname(  
    pid32      pid,          /* ID of process to change */  
    pri16      newprio       /* New priority */  
)
```

menjadi

```
pri16 chname(  
    pid32      pid,          /* ID of process to change */  
    char       *newname,  
    uint32     namelen  
)
```

Modifikasi kode pada chname.c sehingga nama proses bisa diubah bila syscall tersebut dipanggil.

- Hasil setelah diubah pada file: prototypes.h:



```

extern void arp_ntoh(struct arppacket *);
extern void arp_hton(struct arppacket *);

/* in file ascdate.c */
extern status ascdate(uint32, char *);

/* in file bufinit.c */
extern status bufinit(void);

/* in file chprio.c */
extern syscall chprio(pid32, pri16);

/* in file chname.c */
extern pri16 chname(pid32, char*, uint32);

/* in file clkupdate.S */
extern uint32 clkcount(void);

/* in file clkhandler.c */
extern interrupt clkhandler(void);

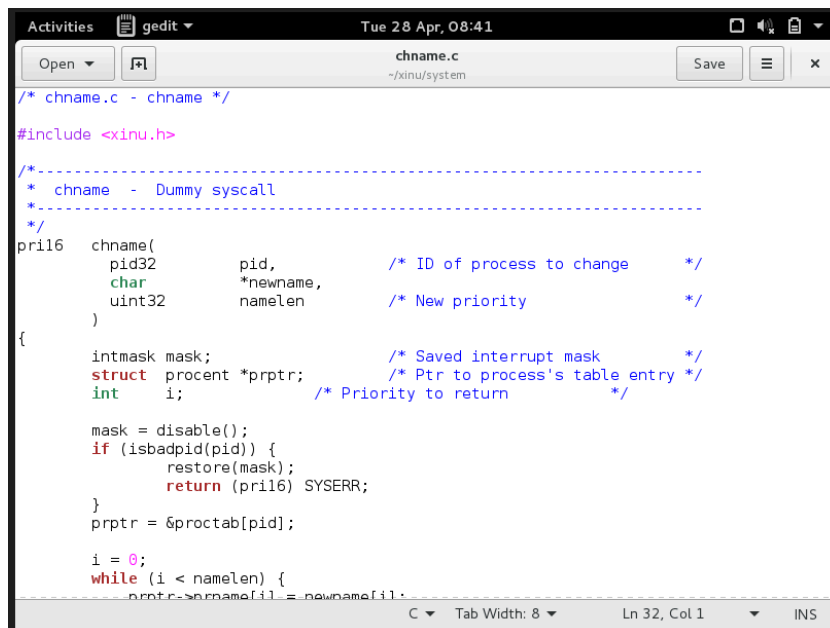
/* in file clkinit.c */
extern void clkinit(void);

/* in file clkdisp.S */
extern void clkdisp(void);

/* in file close.c */

```

- Hasil setelah diubah pada file: chname.c



```

/* chname.c - chname */
#include <xinu.h>

/*-----
 * chname - Dummy syscall
 *-----
 */
pri16 chname(
    pid32    pid,           /* ID of process to change */
    char     *newname,
    uint32    namelen       /* New priority */
)
{
    intmask mask;           /* Saved interrupt mask */
    struct procent *prptr;   /* Ptr to process's table entry */
    int i;                  /* Priority to return */

    mask = disable();
    if (isbadpid(pid)) {
        restore(mask);
        return (pri16) SYSERR;
    }
    prptr = &proctab[pid];

    i = 0;
    while (i < namelen) {
        prptr->name[i] = newname[i];
    }
}

```

```

    i = 0;
    while (i < namelen) {
        prpitr->prname[i] = newname[i];
        i++;
    }
    prpitr->prname[namelen] = '\0';

    restore(mask);
    return oldprio;
}

```

C Tab Width: 8 Ln 22, Col 23 INS

2. Soal 2

Buatlah perintah baru bernama namecmd sesuai dengan langkah-langkah pada no.5 pada modul shell! Berikut adalah kode dalam perintah baru namecmd:

```

printf("My new command");

// pid 3 = netin; mengubah nama proses pid 3 yaitu netin menjadi hello
chname(3, "hello", 5);

return 0;

```

- Menambahkan 'namecmd' pada file shell.c:

```

Activities gedit Tue 28 Apr, 08:55
shell.c ~/xinu/shell

{"sleep", xsh_sleep},
{"tee", xsh_tee},
{"udp", xsh_udpdump},
{"udpecho", xsh_udpecho},
{"udpeserver", xsh_udpeserver},
{"uptime", xsh_uptime},
{"mycmd", xsh_mycmd},
{"namecmd", xsh_namecmd},
{"?", xsh_help}

};

uint32 ncmd = sizeof(cmdtab) / sizeof(struct cmdent);

/*****
/* cmdlookup -- return the index in cmdtab of a command name
/*
*****/

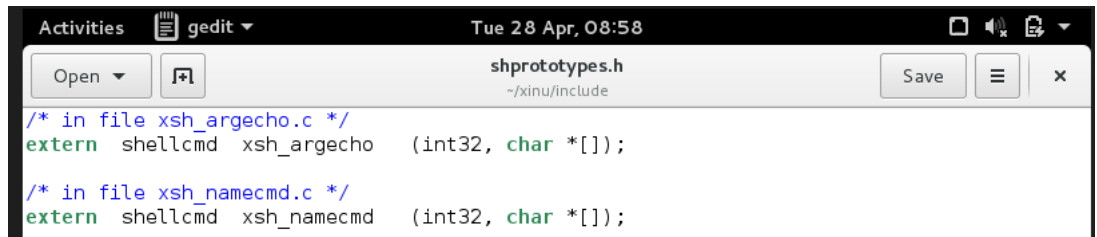
int32 cmdlookup(char *name) {
    int32 indx; /* Index into cmdtab */
    int32 len; /* Length of command name */

    for (indx=0; indx < ncmd; indx++) {
        len = strlen(cmdtab[indx].cname);
        if (strncmp(name, cmdtab[indx].cname, len) == 0) {

```

C Tab Width: 8 Ln 34, Col 36 INS

- Menambahkan 'xsh_namecmd' pada file shprototypes.h:



```
/* in file xsh_argecho.c */
extern shellcmd xsh_argecho (int32, char *[]);

/* in file xsh_namecmd.c */
extern shellcmd xsh_namecmd (int32, char *[]);
```

- Membuat file baru di shell 'xsh_namecmd':



```
/* xsh_namecmd.c - xsh_namecmd */

// Nama : [Zaenarif-Putra-Ainurdin]
// NIM : [2311104049]

#include <xinu.h>
#include <stdio.h>
#include <string.h>

shellcmd xsh_namecmd(int nargs, char *args[]) {
    printf("My new command");
    chname(2, "hello", 5);
    return 0;
}
```

3. Soal 3

Test hasilnya:

- Masuk ke terminal xinu
- Jalankan perintah ps
- Jalankan perintah namecmd
- Jalankan perintah ps
- Lihat nama proses telah berubah
- Berikut hasil berhasil dari semua perintah:

```
Activities Terminal Tue 28 Apr, 09:32
xinu@xinu-develop-end: ~/xinu/compile
File Edit View Search Terminal Help
-----ZAENARIF PUTRA 'AINURDIN-----

Welcome to Xinu!

xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 netin           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005D7FFC 0x005D79AC 8192 0 0
6 ps              curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $ namecmd
My new commandxsh $
xsh $ ps
Pid Name          State Prio Ppid Stack Base Stack Ptr  Stack Size Msg Content
-----
0 prnull          ready  0    0 0x005FDFFC 0x00FFFE4 8192 1 3
1 ipout           wait  500  0 0x005FBFFC 0x005FBF30 8192 0 0
2 hello           wait  500  0 0x005F9FFC 0x005F9ED0 8192 0 0
4 Main process    recv  20   3 0x005E7FFC 0x005E7F34 65536 0 0
5 shell           recv  50   4 0x005D7FFC 0x005D79AC 8192 0 7
8 ps              curr  20   5 0x005F7FFC 0x005F7FC4 8192 0 0

xsh $
CTRL-A Z for help | 115200 8N1 | NOR | Minicom 2.7 | VT102 | Offline | ttyS0
```